

ISC Class XII Physics

Chapter 8 Electromagnetic Waves

10 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Displacement Current
- II. Speed of Electromagnetic Waves in a Medium
- III. Wavelength – Frequency Relation of EM Waves
- IV. Relation between Peak Electric Field (E_0) and Peak Magnetic Field (B_0)
- V. Intensity of an Electromagnetic Wave
- VI. Average Energy Density of Electric and Magnetic Fields

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Formula Sheet

I. Displacement Current

Formula: $I_d = \varepsilon_0 A \frac{dE}{dt}$

I_d = displacement current, ε_0 = permittivity of free space, A = plate area, $\frac{dE}{dt}$ = rate of change of electric field.

II. Electromagnetic Waves

Formula (Speed in a medium): $v = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$

v = speed of EM wave in medium, c = speed of light in vacuum, μ_r = relative permeability, ε_r = relative permittivity.

Formula (Wave equation): $c = f\lambda$

c = speed of light, f = frequency, λ = wavelength.

Formula (E_0 - B_0 relation): $B_0 = \frac{E_0}{c}$

E_0 = peak electric field, B_0 = peak magnetic field, c = speed of light.

Formula (Intensity): $I = \frac{1}{2} \varepsilon_0 c E_0^2$

I = intensity (average power per unit area), ε_0 = permittivity of free space, c = speed of light, E_0 = peak electric field.

Formula (Average energy density – electric): $\langle u_E \rangle = \frac{1}{4} \varepsilon_0 E_0^2$

$\langle u_E \rangle$ = time-averaged electric energy density; equal in magnitude to the average magnetic energy density.

Formula (Average energy density – magnetic): $\langle u_B \rangle = \frac{B_0^2}{4\mu_0} = \langle u_E \rangle$

$\langle u_B \rangle$ = time-averaged magnetic energy density, μ_0 = permeability of free space.

Formula (Total average energy density): $\langle u \rangle = \langle u_E \rangle + \langle u_B \rangle = \frac{1}{2} \varepsilon_0 E_0^2$

$\langle u \rangle$ = total time-averaged energy density of the EM wave (electric + magnetic contributions, which are equal).

I. Displacement Current

Question 1

The rate of change of magnitude of the electric field in the empty space between the plates of a parallel plate capacitor is $4 \times 10^{10} \text{ V m}^{-1}\text{s}^{-1}$. If the area of each plate of the capacitor is 0.01 m^2 , calculate the displacement current.

★ Likely Exam Question

Given: $\frac{dE}{dt} = 4 \times 10^{10} \text{ V m}^{-1}\text{s}^{-1}$, $A = 0.01 \text{ m}^2$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $I_d = \epsilon_0 A \frac{dE}{dt}$

Solution:

Substituting the given values directly into the displacement current formula:

$$I_d = \epsilon_0 A \frac{dE}{dt} = (8.85 \times 10^{-12}) \times (0.01) \times (4 \times 10^{10})$$

$$I_d = 8.85 \times 10^{-12} \times 4 \times 10^8 = 35.4 \times 10^{-4} \text{ A}$$

$$I_d = 3.54 \times 10^{-3} \text{ A}$$

Final Answer

$$I_d = 3.54 \times 10^{-3} \text{ A}$$

Common Mistake: Students often forget to multiply by the plate area A and use only $\epsilon_0 \frac{dE}{dt}$, which gives current density instead of current. Always check that the final unit comes out as amperes (A), not A m^{-2} .

II. Electromagnetic Waves

Question 2

Electromagnetic waves enter a medium of relative permeability 9 and relative permittivity 100. Find the speed of the waves in the medium.

Given: $\mu_r = 9$, $\varepsilon_r = 100$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $v = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$

Solution:

First calculate $\sqrt{\mu_r \varepsilon_r}$:

$$\mu_r \varepsilon_r = 9 \times 100 = 900 \quad \Rightarrow \quad \sqrt{900} = 30$$

Now substitute into the speed formula:

$$v = \frac{c}{\sqrt{\mu_r \varepsilon_r}} = \frac{3.0 \times 10^8}{30}$$

$$v = 1.0 \times 10^7 \text{ m s}^{-1}$$

Final Answer

$$v = 1.0 \times 10^7 \text{ m s}^{-1}$$

Common Mistake: A common slip is multiplying instead of dividing, i.e. writing $v = c\sqrt{\mu_r \varepsilon_r}$. Remember: the speed of a wave in a denser medium is always *less* than c , so the relative factor must divide c .

Question 3

A radar transmitter generates waves of frequency $3 \times 10^9 \text{ Hz}$. What is the wavelength?

Given: $f = 3 \times 10^9 \text{ Hz}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $c = f\lambda \quad \Rightarrow \quad \lambda = \frac{c}{f}$

Solution:

Substituting values:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^9}$$

$$\lambda = 1 \times 10^{-1} \text{ m} = 0.1 \text{ m}$$

Converting to centimetres:

$$\lambda = 10 \text{ cm}$$

Final Answer

$$\lambda = 0.1 \text{ m} = 10 \text{ cm}$$

Common Mistake: Keep track of powers of ten carefully while dividing 10^8 by 10^9 ; a sign error here is a very common source of wrong answers in board exams.

Question 4

A radio station broadcasts at 30 m. Find the frequency of broadcast.

Given: $\lambda = 30 \text{ m}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $c = f\lambda \Rightarrow f = \frac{c}{\lambda}$

Solution:

Substituting values:

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{30}$$

$$f = 1 \times 10^7 \text{ Hz}$$

Converting to megahertz:

$$f = 10 \text{ MHz}$$

Final Answer

$$f = 1 \times 10^7 \text{ Hz} = 10 \text{ MHz}$$

Common Mistake: Do not confuse MHz (10^6 Hz) with GHz (10^9 Hz) while converting the final answer – this is a frequent unit-conversion trap.

Question 5

A plane electromagnetic wave has a maximum electric field of $3.20 \times 10^{-4} \text{ V m}^{-1}$. What is the maximum magnetic field?

★ Likely Exam Question

Given: $E_0 = 3.20 \times 10^{-4} \text{ V m}^{-1}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $B_0 = \frac{E_0}{c}$

Solution:

Substituting values:

$$B_0 = \frac{E_0}{c} = \frac{3.20 \times 10^{-4}}{3.0 \times 10^8}$$

$$B_0 = 1.067 \times 10^{-12} \text{ T}$$

$$B_0 \approx 1.07 \times 10^{-12} \text{ T}$$

Final Answer

$$B_0 = 1.07 \times 10^{-12} \text{ T}$$

Common Mistake: Students sometimes multiply E_0 by c instead of dividing, forgetting that B_0 is always much smaller in magnitude than E_0 (SI units), since c is a very large number in the denominator.

Question 6

A laser beam has intensity $2.5 \times 10^{14} \text{ W m}^{-2}$. Find the amplitudes of electric and magnetic fields in the beam.

★ Likely Exam Question

Given: $I = 2.5 \times 10^{14} \text{ W m}^{-2}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $I = \frac{1}{2}\epsilon_0 c E_0^2 \Rightarrow E_0 = \sqrt{\frac{2I}{\epsilon_0 c}}, \quad B_0 = \frac{E_0}{c}$

Solution:

Calculate the denominator first:

$$\epsilon_0 c = (8.85 \times 10^{-12})(3.0 \times 10^8) = 2.655 \times 10^{-3}$$

Now find E_0 :

$$E_0 = \sqrt{\frac{2I}{\epsilon_0 c}} = \sqrt{\frac{2 \times 2.5 \times 10^{14}}{2.655 \times 10^{-3}}} = \sqrt{1.883 \times 10^{17}}$$

$$E_0 = 4.34 \times 10^8 \text{ N C}^{-1}$$

Now find B_0 using $B_0 = E_0/c$:

$$B_0 = \frac{4.34 \times 10^8}{3.0 \times 10^8} = 1.44 \text{ T}$$

Final Answer

$$E_0 = 4.34 \times 10^8 \text{ N C}^{-1}, \quad B_0 = 1.44 \text{ T}$$

Common Mistake: The intensity formula has a factor of $\frac{1}{2}$ because E_0 is the *peak*, not the rms, value. Forgetting this factor doubles the intensity and gives a wrong (larger) value of E_0 .

Question 7

In a plane EM wave, the magnetic field oscillates sinusoidally with a frequency of 2.0×10^{10} Hz and an amplitude of 46 Wb m^{-2} . Find the wavelength of the wave and the amplitude of the oscillating electric field.

Given: $f = 2.0 \times 10^{10}$ Hz, $B_0 = 46 \text{ Wb m}^{-2}(\text{T})$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $\lambda = \frac{c}{f}$, $E_0 = B_0 c$

Solution:

Wavelength:

$$\lambda = \frac{c}{f} = \frac{3.0 \times 10^8}{2.0 \times 10^{10}} = 1.5 \times 10^{-2} \text{ m}$$

Amplitude of electric field:

$$E_0 = B_0 c = 46 \times 3.0 \times 10^8 = 138 \times 10^8 \text{ V m}^{-1}$$

Final Answer

$$\lambda = 1.5 \times 10^{-2} \text{ m}, \quad E_0 = 138 \times 10^8 \text{ V m}^{-1}$$

Common Mistake: When B_0 is given, always use $E_0 = B_0 c$ (not $E_0 = B_0/c$). Mixing up the direction of this relation is the most common error in such problems.

Question 8

In a plane EM wave, the electric field oscillates sinusoidally with a frequency of 2×10^{10} Hz and an amplitude of 48 V m^{-1} . Find the wavelength and the amplitude of the oscillating magnetic field.

Given: $f = 2 \times 10^{10}$ Hz, $E_0 = 48 \text{ V m}^{-1}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $\lambda = \frac{c}{f}$, $B_0 = \frac{E_0}{c}$

Solution:

Wavelength:

$$\lambda = \frac{c}{f} = \frac{3.0 \times 10^8}{2 \times 10^{10}} = 1.5 \times 10^{-2} \text{ m}$$

Amplitude of magnetic field:

$$B_0 = \frac{E_0}{c} = \frac{48}{3.0 \times 10^8} = 16 \times 10^{-8} \text{ T}$$

$$B_0 = 0.16 \times 10^{-6} \text{ T} = 0.16 \mu\text{T}$$

Final Answer

$$\lambda = 1.5 \times 10^{-2} \text{ m}, \quad B_0 = 0.16 \text{ } \mu\text{T}$$

Common Mistake: Do not confuse this question with Question 7 – here E_0 is given and B_0 must be found using $B_0 = E_0/c$ (division), the reverse of the previous case.

Question 9

In a plane electromagnetic wave, the electric field varies with time having an amplitude of 1.0 V m^{-1} . Find the average energy density of the magnetic field and that of the electric field.

★ **Likely Exam Question**

Given: $E_0 = 1.0 \text{ V m}^{-1}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $\langle u_E \rangle = \frac{1}{4} \epsilon_0 E_0^2$, $\langle u_B \rangle = \langle u_E \rangle$

Solution:

Calculate the average electric energy density:

$$\langle u_E \rangle = \frac{1}{4} \epsilon_0 E_0^2 = \frac{1}{4} (8.85 \times 10^{-12}) (1.0)^2$$

$$\langle u_E \rangle = 2.2125 \times 10^{-12} \text{ J m}^{-3} \approx 2.2 \times 10^{-12} \text{ J m}^{-3}$$

Since the average electric and magnetic energy densities of an EM wave are always equal:

$$\langle u_B \rangle = \langle u_E \rangle = 2.2 \times 10^{-12} \text{ J m}^{-3}$$

Final Answer

$$\langle u_E \rangle = \langle u_B \rangle = 2.2 \times 10^{-12} \text{ J m}^{-3} \text{ (each)}$$

Common Mistake: A very common conceptual error is assuming the magnetic energy density is negligible compared to the electric one. In an EM wave, the time-averaged electric and magnetic energy densities are always exactly equal.

Question 10

In a plane electromagnetic wave, the amplitude of the magnetic field is $5.0 \times 10^{-6} \text{ T}$. Find the amplitude of the electric field and the total average energy density of the wave.

★ **Likely Exam Question**

Given: $B_0 = 5.0 \times 10^{-6} \text{ T}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $E_0 = B_0 c$, $\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2$

Solution:

Amplitude of electric field:

$$E_0 = B_0 c = (5.0 \times 10^{-6})(3.0 \times 10^8) = 1.5 \times 10^3 \text{ V m}^{-1}$$

Total average energy density (electric + magnetic contributions):

$$\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2 = \frac{1}{2} (8.85 \times 10^{-12}) (1.5 \times 10^3)^2$$

$$\langle u \rangle = \frac{1}{2} (8.85 \times 10^{-12}) (2.25 \times 10^6)$$

$$\langle u \rangle = \frac{1}{2} (1.99 \times 10^{-5}) = 1.0 \times 10^{-5} \text{ J m}^{-3}$$

Final Answer

$$E_0 = 1.5 \times 10^3 \text{ V m}^{-1}, \quad \langle u \rangle = 1.0 \times 10^{-5} \text{ J m}^{-3}$$

Common Mistake: For *total* average energy density use the factor $\frac{1}{2} \epsilon_0 E_0^2$ (electric + magnetic parts combined). Using $\frac{1}{4} \epsilon_0 E_0^2$ here gives only the electric-field contribution, which is half the correct total.